

MID TERM EXAMINATION
B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
 3rd Semester, November, 2023

Paper Code: CIC 211**Subject: Object-Oriented Programming using C++****Time: 1:30 Hrs.****Max Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1	(a) Explain inline function with example.	2	CO1
	(b) Explain the concept of array of objects and pointers to objects with an example.	3	CO2
	(c) What is this pointer?	2	CO1
	(d) Discuss the importance of Data Abstraction and Encapsulation with a real time example.	3	CO1
2	(a) Design a class from which only one object can be created. If more than one object is created the program should terminate.	4	CO2
	(b) Name the operators that cannot be overloaded. Justify why they cannot be overloaded? Write program to overload << and >> operators in complex numbers.	6	CO3
3	(a) Display a stack program in C++ using class, overloaded constructors, member functions (push, pop), friend function display stack content.	7	CO2
	(b) Differentiate between run time polymorphism and compile time polymorphism.	3	CO2
4	(a) Explain default, parameterized constructor and copy constructor with an example?	5	CO3
	(b) Define dynamic memory allocation. Write a C++ program to illustrate new and delete operators. Explain why new is better than malloc in C.	5	CO2, CO3